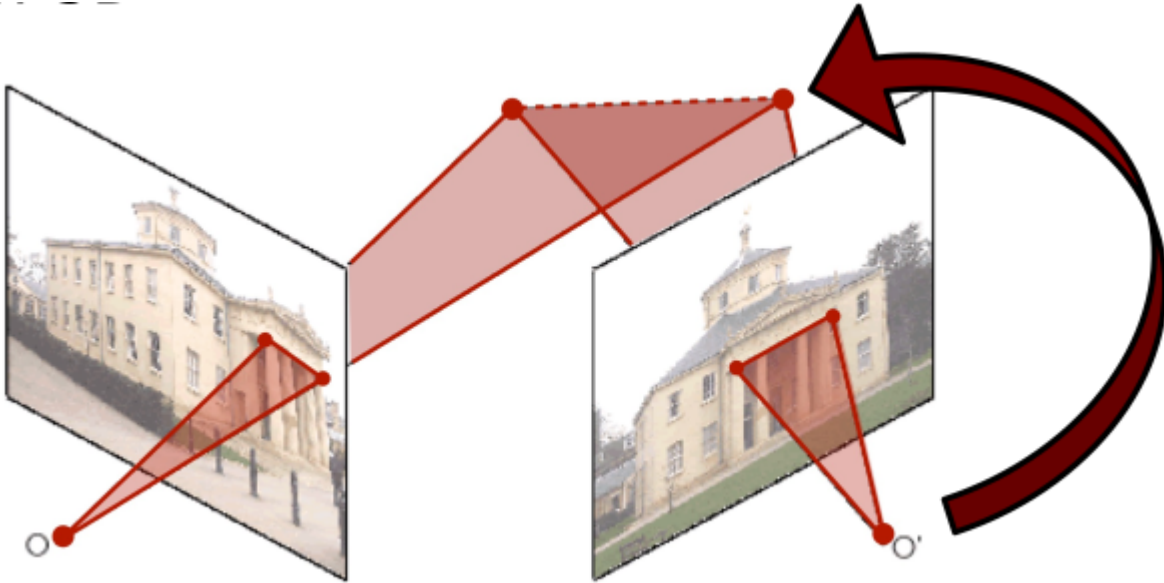
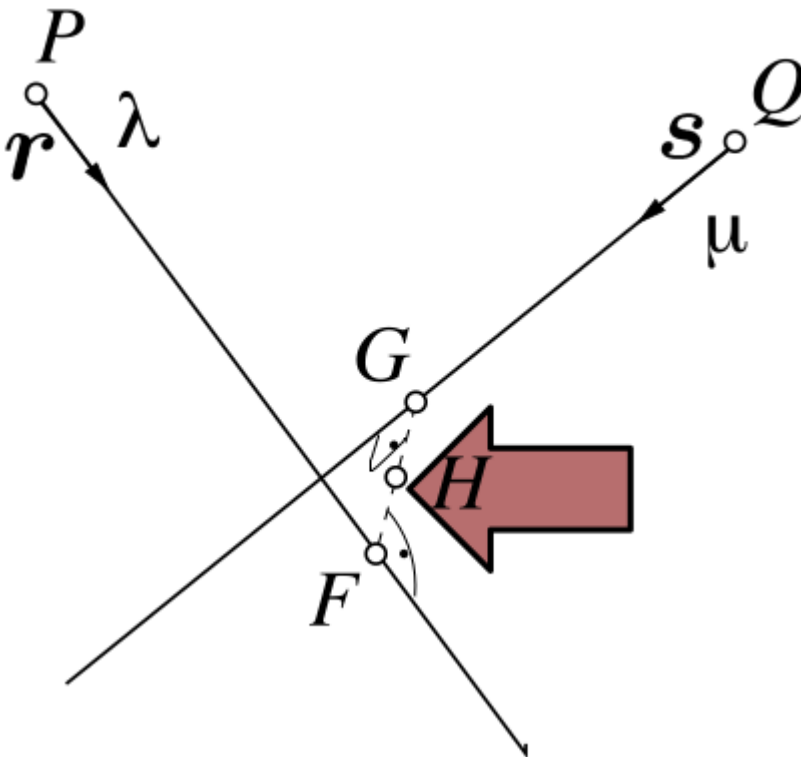


L09: Stereo Triangulation

- Given relative orientation of 2 images, compute points in 3D



- Lines may not intersect in 3D



Geometric Solution

- 2 lines in 3D
 - $f = q + \lambda r$ and $g = p + \mu s$
 - Points: $p = X_{O'}$, $q = X_{O''}$
 - and the directions (calibrated camera)
 - $r = R^{Tk} x'$ and $s = R''^{Tk} x''$

- With ${}^k x' = (x', y', c)^T$ and ${}^k x'' = (x'', y'', c)^T$

Constraint

- Shortest connection requires orthogonality of line FG

$$\begin{aligned} (f - g) \cdot r &= 0 & (f - g) \cdot s &= 0 \\ (p + \lambda r - q - \mu s) \cdot r &= 0 & (p + \lambda r - q - \mu s) \cdot s &= 0 \\ (X_{O'} + \lambda r - X_{O''} - \mu s)^T \cdot r &= 0 & (X_{O'} + \lambda r - X_{O''} - \mu s)^T \cdot s &= 0 \end{aligned}$$

- Solving equation for λ, μ
 - Yield directly F and G
- We compute H as the middle of the line connecting the points F and G
- Rewrite previous equation and put into matrix:

$$\begin{bmatrix} r^T r & -s^T r \\ r^T s & -s^T s \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \end{bmatrix} = \begin{bmatrix} (X_{O''} - X_{O'})^T r \\ (X_{O''} - X_{O'})^T s \end{bmatrix}$$

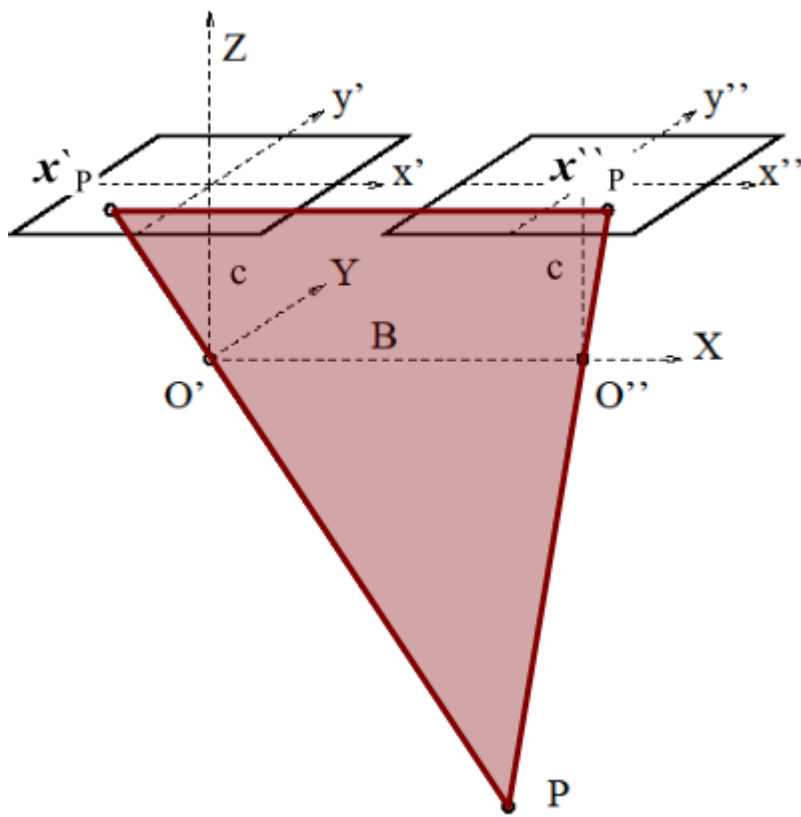
$$Ax = b$$

- The 3D point H is the middle of the line connecting F and G
 - λ, μ directly yield F and G

$$H = \frac{F + G}{2}$$

- **Downside:**
 - No accountability for uncertainties
 - Not statistically optimal

Stereo Normal Case



1. Z-coordinate from intercept theorem

$$Z = c \frac{B}{-(x'' - x')}$$

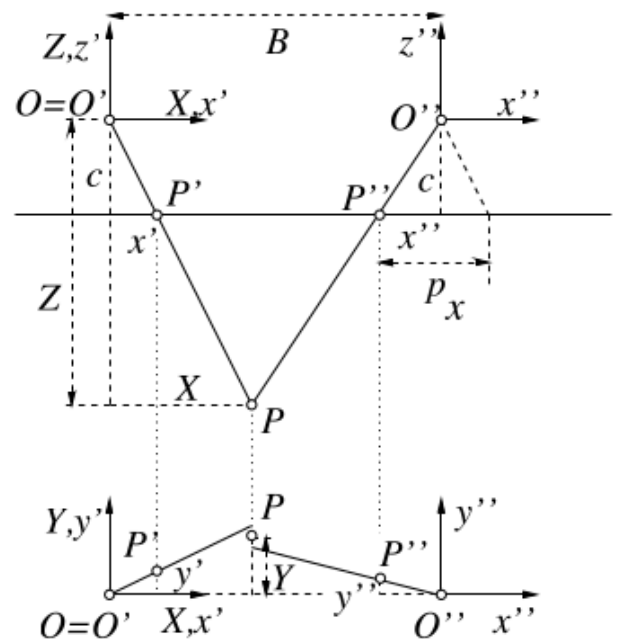
2. X-coordinate

$$X = x' \frac{B}{-(x'' - x')}$$

3. Y-coordinate by mean

$$\frac{Y}{X} = \frac{\frac{y' + y''}{2}}{x'}$$

$$\Rightarrow Y = \frac{y' + y''}{2} \frac{B}{-(x'' - x')}$$



Intersection of 2 Rays for the Stereo Normal Case

- Yields the Point Coordinates
- Parallax also called **disparity**
- x-parallax $p_x = x'' - x'$ corresponds to depth Z
 - is key element
- y-parallax $p_y = y'' - y'$
 - corresponds to **consistency** of image points

- Should be small (Stereo normal case)

If y-parallax: $p_y = y'' - y' = 0$

- $X = x' \frac{B}{-p_x} \quad Y = y' \frac{B}{-p_x} \quad Z = c \frac{B}{-p_x}$

- Rewrite:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} -\frac{B}{p_x} & 0 & 0 \\ 0 & -\frac{B}{p_x} & 0 \\ 0 & 0 & -\frac{B}{p_x} \end{bmatrix} \begin{bmatrix} x' \\ y' \\ c \end{bmatrix}$$

- As homogeneous coordinates

$$\begin{bmatrix} U \\ V \\ W \\ T \end{bmatrix} = \begin{bmatrix} B & 0 & 0 & 0 \\ 0 & B & 0 & 0 \\ 0 & 0 & Bc & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \\ p_x \end{bmatrix}$$

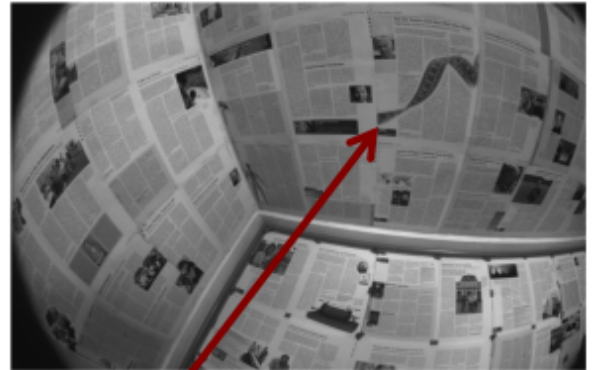
camera-dependent point-dependent

- For a set of points:

- $\{x', y'\}$ (1st image)

- **Parallax map:** $\{x', y', p_x\}$

- directly leads to the 3D coordinates of the point



parallax map $\{x', y', p_x\}_1 \dots \{x', y', p_x\}_N$



parallax map $\{x', y', p_x\}_1 \dots \{x', y', p_x\}_N$

Absolute Orientation

Relative Orientation

- 3D mapping result of relative orientation:
 - photogrammetric model
 - contains
 - Parameters of relative orientation of both cameras
 - 3D coordinates of N points in a local coordinate frame m
 - models only known up to a similarity transform (for calibrated cameras)

Absolute Orientation

- Similarity transform maps photogrammetric model m into the object reference frame o

$${}^oX_n = \lambda R^m X_n + T$$

- 7 DoF for similarity transform
 - (3 rotation, 3 translation, 1 scale)
- **Control points** are required

Least Squares Solution

- Non-linear least squares solution to compute the parameters
- At least **3 known control points** with (X,Y,Z)
- Approach similar to the ICP algorithm
 - ICP typically assumes a known scale of 1
- Find similarity transform

$$\bar{x}_n = \lambda R x_n + t$$

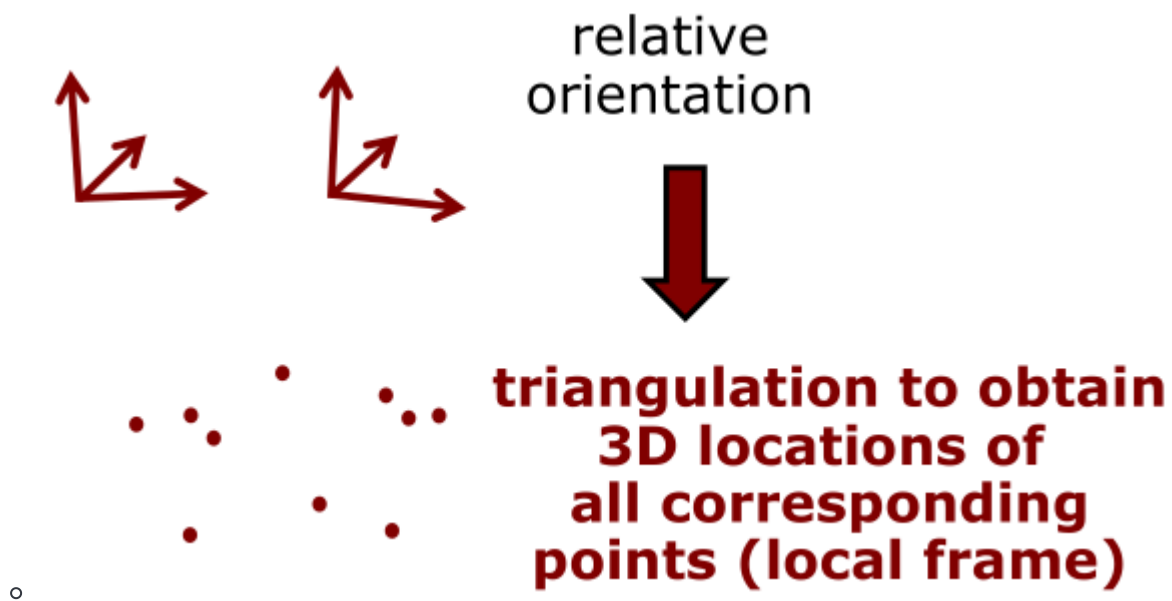
- transforms a set of points $\{\mathbf{x}_n\}$
- $\{\mathbf{x}_n\}$ as close as possible to point set $\{\mathbf{y}_n\}$
- Known correspondences
- Minimise the squared error

$$\sum \|\mathbf{y}_n - \bar{\mathbf{x}}_n\|^2 \rightarrow \min$$

2-Step Solution

1. Relative orientation

- without control points and 3D
- location of corresponding points in a local frame



2. Absolute orientation of cameras

- corresponding points through control points

